

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000831	Version:	5.0, CURRENT
	Status:	Effective	Effective Date:	19-Jul-2011
Title:	SM DG57-DG517. Syringe-dilutor replacement.			Page 2 of 9

DG-57 / DG-517 Service Manual

SYRINGE / DILUTOR REPLACEMENT

Procedure description
This procedure describes how replace the syringe and/or the dilutor. Note: Document REGD-0000831_v4 replaces document REGD-0000831_v3.

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EQUIPMENT

1. An instrument connected with a PC with the software installed.
2. A new syringe code 231821 or a new Cavro Dilutor XC3000 (with valve) code 232228.
3. 2 stick tie wraps holders.

SET UP

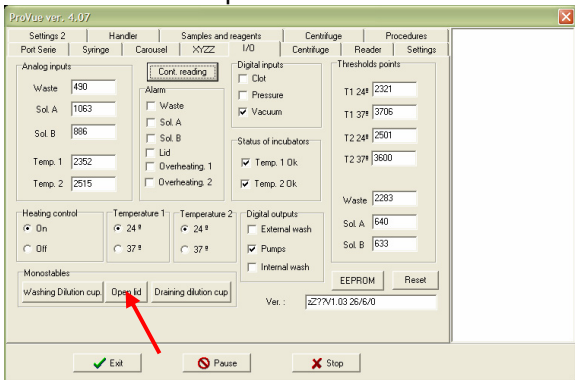
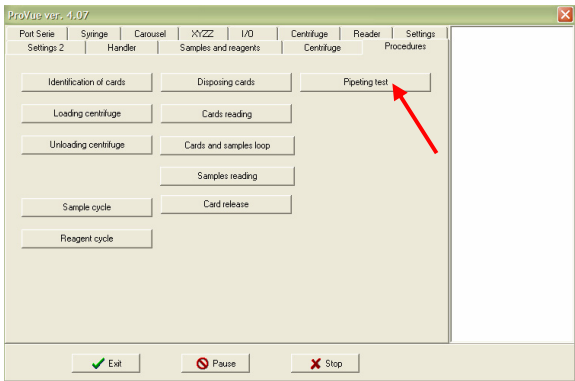
1. Turn on the instrument.
2. Start the ST software (Technical Service software).
3. Enter the Technical Service password (d+g_diana).

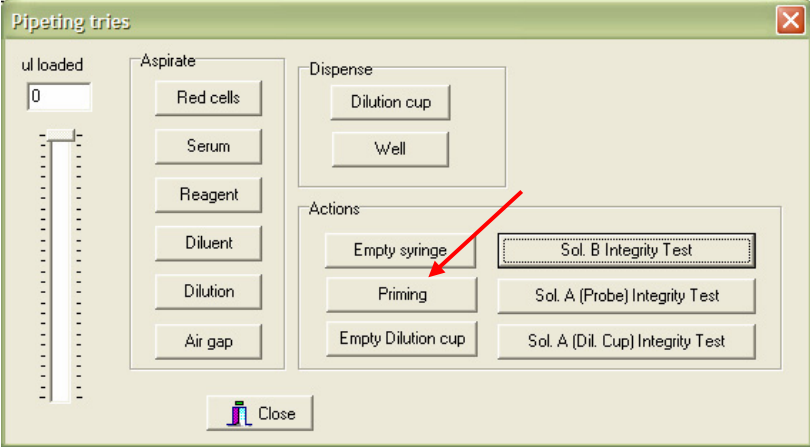
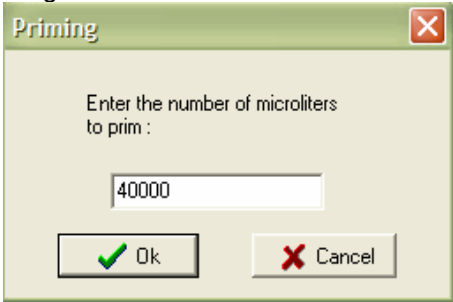
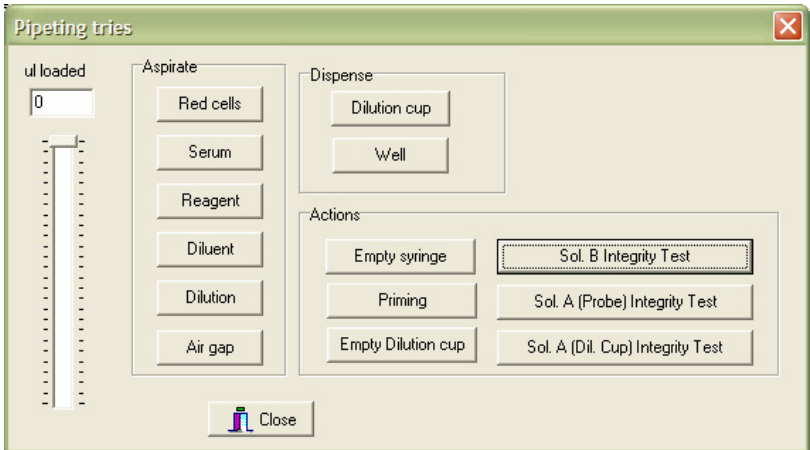


WARNING!

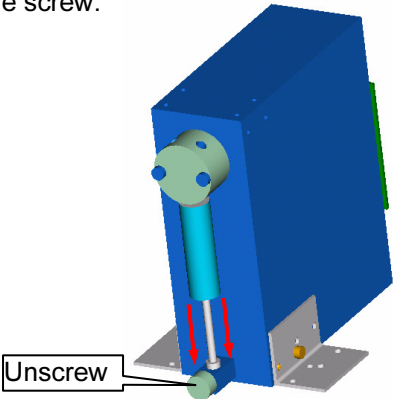
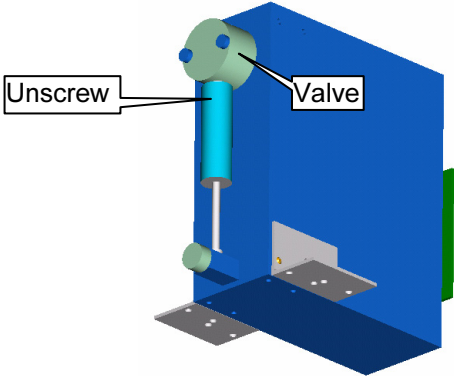
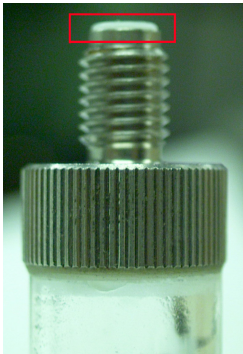
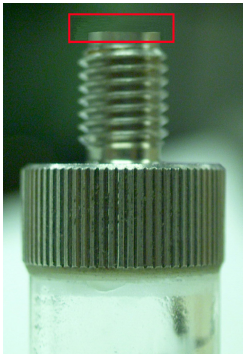
There may be rests of blood in the syringe. Perform decontamination procedure REGD-0000787 before changing the syringe.

PROCEDURE

Process	Expected Results
1. Click on the "I/O" tab. Press on "Open lid" button.  Picture 1 Open the front bottles door. Disconnect and remove the "Solution A" bottle and scoop all the liquid. place again the bottle in the instrument and reconnect the racors. Close the front bottles door. 2. Click on the "Procedures" tab. Press on "Pipeting test" button.  Picture 2 3. Click on "Priming" button.	The door opens. New window appears (Picture 3) New window

Process	Expected Results
 Picture 3 4. Write “40000” in “Priming” window.  Picture 4 Wait until the software ends the procedure. 5. Click on “Close” button.  Picture 5 6. REPLACING THE SYRINGE: Don’t make step 6 to 9 if dilutor is to be replaced. To establish whether the syringe has to be replaced it must be uninstalled. To uninstall the syringe move the syringe plunger downwards by selecting ‘air gap’ inside ‘aspirate’ square (picture 5) and then choose ‘2000’.	appears (see Picture 4) --- --- ---

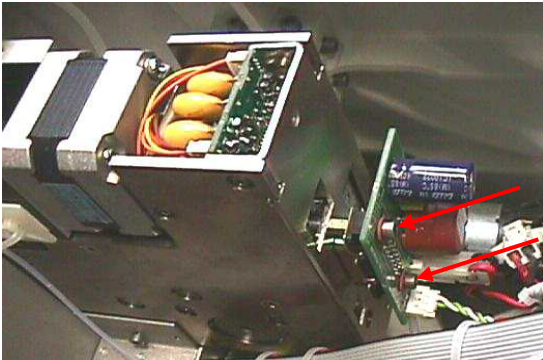
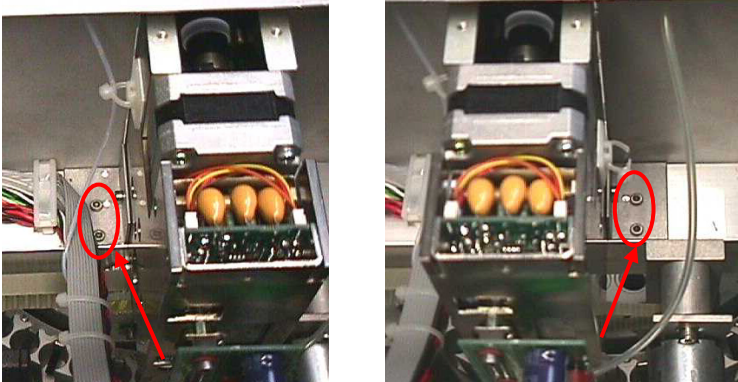
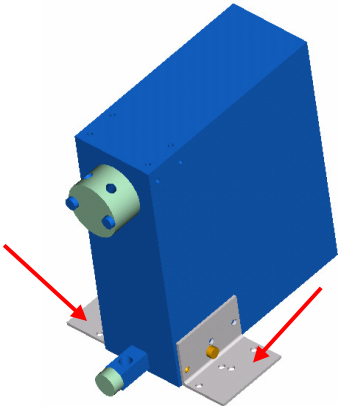
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Process	Expected Results
After plunger has move down unscrew manually the lock screw of the piston. Don't extract completely the screw.  Picture 6 7. Unscrew manually the glass into the valve.  Picture 7 Remove the glass and the piston from the instrument. 8. Check that the teflon insert still protrudes above the front screw (Picture 8). If the Teflon insert is not visible looking at the syringe front screw directly from the side (Picture 9), the syringe should be replaced.  Picture 8  Picture 9	The syringe will be removed from the dilutor.

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Process	Expected Results
9. If necessary replace the new syringe screwing it in the valve: It is recommended that syringe would be tightened between 1/8th and 1/4th, with a minimum of 1/16th of turn, after it touches-off. Finally screw the lock screw in the dilutor (See pictures 8 and 9). User should also be aware that because of compression on the Teflon insert, the touch-off point will change (slightly) each time the syringe is re-tightened. 10. REPLACING THE DILUTOR. Unscrew, manually, the tubes connections in the dilutor valve. Picture 10 Be careful to not loose the ring of the left connector. Picture 11 11. Cut the tie wraps of the tubes. Picture 12	- - - - - - - - -

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Process	Expected Results
12. Remove the power supply & communication PCB unscrewing the fixing screws.  Picture 13 13. Remove the dilutor unscrewing the support screws from holders of dilutor.  Picture 14 14. Remove the holders from the old dilutor and put them the new dilutor.  Picture 15	--- The dilutor can be removed. ---

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DOCUMENT CHANGE HISTORY

Document change history			
Version	Date	Change	Rationale
3.0	2005/05/09	- Document REGD-0000831_v3 replaces document 57058V0M001. - Added jumpers configuration in step 16, syringe replacement criteria in step 7 and screwing recommendation in step 9.	Uploaded document to GxPharma.
4.0	2011/04/28	- Updated pictures 1, 2, 3, 4 and 5.	Updated pictures due to the new fluidic controls implemented since software versions 3.x.
4.0	2011/04/28	- New part number 232228. - Updated picture 17 with the new Cavro XC3000 selector configuration. - Eliminated from step 16 the jumpers configuration.	Updated information related with the new Cavro Dilutor XC3000.
4.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY/MM/DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
4.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
5.0	See document header.	Document REGD-0000831_v4 replaces document REGD-0000831_v3.	Document version changed.
5.0	See document header.	Step 1 modified.	Only solution A bottle must be emptied.
5.0	See document header.	Steps 6 to 8 explanation modified.	Ensure proper syringe uninstallation.
5.0	See document header.	Step 18 an 19 upgraded upgraded.	Ensure proper syringe and racords installation.
5.0	See document header.	Document title.	Commercial reference substituted by project reference.
5.0	See document header.	Document header.	Increase information to easy document search.